1 Numerical Experiments

This chapter describes the experimental framework used in the repository for randomized least-squares solvers and for the later covariance-transport problem. The goal is not only to illustrate the theory, but to test whether the geometric statements from ??→p.?? and the perturbation viewpoint from ??→p.?? remain visible in computed residuals, computed solutions, covariance approximations, and transport errors. The experiments also provide the first controlled setting in which the floating-point questions from ??→p.?? can be studied in the same benchmark.

The experiments use intentionally synthetic data. This is useful: by generating the data matrix, covariance structure, and reference solution directly, we separate the effect of the sketch from the complications of a downstream application. The experiments are therefore simple enough to interpret and still rich enough to show the interaction between embedding quality, conditioning, regularization, and arithmetic.

1.1 Experimental Objective

The experiments are naturally divided into two parts. Part I studies sketched least-squares solvers as the first model problem. Part II studies randomized covariance transport as the climate-facing operator problem introduced in ??→p.??. The experiments currently cover mainly the first part, but the second part already has a clear mathematical specification.

For the least-squares experiments, the main question is how the quality of a sketched solve changes as the sketch dimension varies and as the sketch family changes. The experiments track four quantities:

- the mean absolute error of the averaged sketch solution,
- the residual ratio relative to a reference least-squares solve,
- the relative solution error with respect to the reference solution,
- the condition number of the sketched matrix SA.

These quantities reflect different numerical phenomena. The residual ratio measures how well the sketched problem preserves the least-squares objective. The solution error is more sensitive to conditioning and therefore detects failures that may not be visible at the residual level. The condition number of SA is not an error measure by itself, but it often explains why two sketches with similar residual behavior may differ strongly at the level of the computed solution.

The experiments are therefore not only comparisons of random projections. They test how approximation quality, solver sensitivity, and arithmetic interact in one concrete computational pipeline.

1.2 Synthetic Least-Squares Model

The implementations use one common synthetic model. A matrix $A \in \mathbb{R}^{n\times d}$ is generated with independent standard normal entries, and the reference solution is chosen as

$$x_\star = \mathbf{1} \in \mathbb{R}^d.$$

The right-hand side is then formed as

$$b = Ax_\star + e,$$

where e is either zero or a Gaussian perturbation scaled by a prescribed noise level. In this way, both the consistent and noisy regimes are available within the same template.

For each generated instance, a reference least-squares solution x_{ref} is computed from the full system. In the R implementation this is done with `qr.solve`, while the Octave prototypes use the backslash solver. The reference residual

$$r_{\mathrm{ref}} = Ax_{\mathrm{ref}} - b$$

provides the baseline against which the sketched residuals are compared. This is important because each trial is measured against the corresponding unsketched least-squares solve.

The synthetic model is deliberately modest. It does not attempt to mimic the spectral structure of a climate-data operator or the coherence patterns of an application matrix. Its role is to isolate the basic mechanisms of randomized least squares under controlled conditions.

1.3 Implemented Sketch Families

The experiments support three sketch families in the main R package: Gaussian sketches, Rademacher sketches, and the SRHT. The Octave prototypes currently cover Gaussian sketches and the SRHT. This reflects the current state of the software rather than a conceptual distinction. The R package is the intended long-term implementation base, while the Octave scripts remain useful as compact prototypes.

The Gaussian sketch is the cleanest theoretical baseline. The Rademacher sketch replaces Gaussian entries by random signs and tests how much of the same behavior survives under a simpler discrete distribution. The SRHT introduces structure and therefore changes both the cost model and the embedding constants. It is the first sketch family in the project for which it is natural to ask whether the additional algebraic structure is worth the analytical complexity.

There is one practical restriction that matters experimentally. The current SRHT implementations require the row dimension to be a power of two. Therefore, the R and Octave code round n upward when needed before constructing the Hadamard-based sketch. This is a simple example of a theoretical assumption becoming an implementation condition.

1.4 Per-Trial Computation

For a fixed sketch size s, one trial proceeds as follows. A sketching matrix $S \in \mathbb{R}^{s\times n}$ is drawn, the compressed system

$$SAx \approx Sb$$

is formed, and the reduced least-squares problem is solved. The computed solution for that trial is denoted by $\hat{x}$. The experiment then records:

$$\begin{aligned}
\text{residual ratio} &= \frac{\|A\hat{x} - b\|_2}{\|Ax_{\text{ref}} - b\|_2}, \\
\text{solution error} &= \frac{\|\hat{x} - x_{\text{ref}}\|_2}{\|x_{\text{ref}}\|_2}, \\
\text{condition number} &= \kappa_2(SA).
\end{aligned}$$

These definitions are aligned with the theoretical discussion in ??→p.??. The residual ratio corresponds to the sketch-and-solve guarantee that the compressed problem should preserve the least-squares objective up to a controlled distortion. The solution error reflects the fact that preserving the residual does not automatically guarantee an equally small perturbation in the minimizer. The condition number of SA is recorded because it is one of the main mechanisms that links these two behaviors.

In addition, the implementations average the independent trials at each sketch size. If $\hat{x}^{(1)}, \ldots, \hat{x}^{(m)}$ are the trial solutions, the code forms their mean and reports the mean absolute error of this averaged estimator relative to the planted vector $x_\star$. This is not a standard RandNLA performance criterion, but it is useful here because it exposes the variability of the individual trial solutions.

1.5 Parameter Sweeps

The experiments vary the sketch dimension s over a prescribed range and repeat the randomized solve several times for each value. The main routine `sketch_lstsq_experiment` accepts the parameters n, d, a vector of sketch sizes, a trial count, a noise level, and the sketch family. Specialized wrappers expose

Gaussian and SRHT runs directly, but the underlying routine already supports all three sketch families.

The Octave scripts implement the same logic in a more direct form. The Gaussian prototype sweeps over `s_vals = 1:99` by default for a problem of size $n = 100$, $d = 10$, while the SRHT prototype uses a coarser sweep adapted to the power-of-two requirement. In both cases, several independent sketches are drawn for each s, and the recorded history stores the sketch size together with the averaged error metrics.

This repeated-trial design is essential. A single randomized sketch may be atypical, especially for small sketch sizes. Averaging over several trials does not remove randomness from the method, but it makes the observed trends much more interpretable.

1.6 Precision-Sensitivity Baseline

The mixed-precision part of the experimental framework is under active development, but the current Octave Gaussian experiment already contains a useful first comparison. It can run either in ordinary double precision or, when the symbolic package is available, in a higher-precision mode used as a surrogate for reduced rounding error. This is not a full mixed-precision implementation in the sense discussed in ??→p.??. It is a controlled way to ask whether a visible loss of accuracy is mainly geometric or mainly arithmetic.

That distinction matters. If increasing arithmetic precision leaves the curves essentially unchanged, the dominant limitation is likely the sketch itself or the conditioning of the reduced problem. If the higher-precision run materially improves the behavior, then floating-point perturbations are already visible at this synthetic scale. The current precision experiment is therefore best viewed as a diagnostic baseline for the later mixed-precision work.

1.7 Least-Squares Sketch-Size Sweep

The first implemented experiment tests how the quality of a sketched solve changes as the sketch dimension increases. The setup follows the synthetic model from Section 1.2→p.2 with $n = 1024$, $d = 20$, a planted solution $x_\star = \mathbf{1}$, and additive Gaussian noise scaled to a signal-to-noise ratio of approximately 10^4, so that the reference residual is nonzero and the residual ratio is a meaningful diagnostic. A Gaussian sketch is used, the sketch dimension s is swept over the values

$$s \in \{30, 40, 60, 80, 100, 150, 200, 300\},$$

and 20 independent trials are drawn for each value.

Figure 1.1→p.5 shows the mean residual ratio as a function of s. The expected qualitative behaviour is clearly visible: the residual ratio decreases toward one

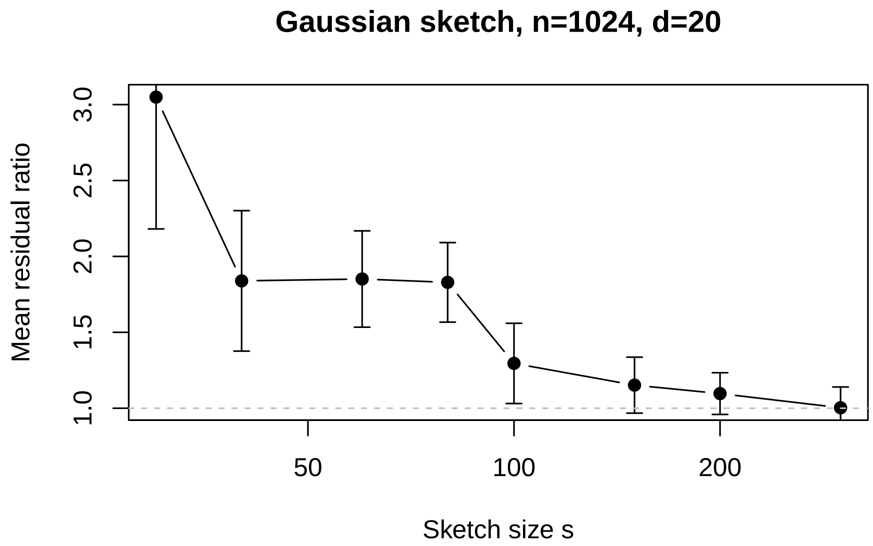

Figure 1.1: Mean residual ratio versus sketch size for Gaussian sketching of a noisy least-squares problem with $n = 1024$, $d = 20$. Error bars indicate one standard deviation across 20 independent trials.

as the sketch dimension grows, with the most rapid improvement occurring for s between two and five times the column dimension. The error bars (one standard deviation across trials) shrink as s grows, reflecting the reduced variability of the embedding at larger dimensions.

Table 1.2 →p.6 reports the mean residual ratio and mean relative solution error for each sketch size. Two observations are worth highlighting. First, the residual ratio approaches one much more slowly than the solution error approaches machine precision: the residual ratio is still approximately 1.15 at $s = 150$, whereas the solution error is already at the 10^{-16} level. This confirms the theoretical expectation that residual preservation and solution preservation are different diagnostics, and that the solution error is governed mainly by the conditioning of SA rather than by the residual quality alone. Second, the solution error still improves with s even though the conditioning of SA may vary, because the overall perturbation scale decreases with the sketch dimension.

The main qualitative expectation is straightforward. As the sketch dimension grows, the residual ratio should approach one, because the compressed problem should preserve the geometry of the original least-squares problem more and more accurately. The solution error should also generally improve, but often less monotonically and sometimes more slowly, because it depends more strongly on the conditioning of the sketched operator. Therefore, the condition number $\kappa_2(SA)$ should be read together with the error curves.

This is especially important when comparing dense and structured sketches. A structured sketch may be attractive because it is cheaper to apply, and yet it may

s	Mean residual ratio	Mean rel. solution error
30	3.049	1.18×10^{-15}
40	1.839	7.40×10^{-16}
60	1.851	7.56×10^{-16}
80	1.829	7.37×10^{-16}
100	1.295	5.64×10^{-16}
150	1.152	5.20×10^{-16}
200	1.096	4.88×10^{-16}
300	1.003	4.79×10^{-16}

Table 1.2: Least-squares sketch-size sweep: Gaussian sketch, $n = 1024$, $d = 20$, 20 trials per sketch size. Residual ratio is $\|A\hat{x} - b\|_2 / \|Ax_{\mathrm{ref}} - b\|_2$; solution error is $\|\hat{x} - x_{\mathrm{ref}}\|_2 / \|x_{\mathrm{ref}}\|_2$.

produce larger fluctuations or require a larger sketch size before the residual and solution metrics stabilize. Conversely, a dense Gaussian sketch may look cleaner numerically while being less attractive computationally. The experiments are designed to make this tradeoff visible.

This chapter is therefore best read as a methodology and interpretation chapter. The benchmark structure is already concrete and implemented for least squares, and the mixed-precision side is still being expanded. Even so, the framework is already explicit enough to support the main thesis question: whether randomized compression and altered arithmetic can be studied in one coherent numerical-analysis picture.

1.8 Covariance-Transport Experiment Program

The second experimental layer of the thesis tests the theory from ??→p.?? just as directly as the least-squares benchmarks test the embedding theory from ??→p.??. Instead of tracking only the compressed least-squares residual, one now measures the quality of a randomized low-rank covariance approximation and the perturbation it induces in the regularized transport map.

For a covariance operator C and its computed approximation $\widehat{C}$, the basic numerical diagnostics should include

$$\|C - \widehat{C}\|_2, \qquad \|T_\lambda - \widehat{T}_\lambda\|_2, \qquad \|Z_f T_\lambda - Z_f \widehat{T}_\lambda\|_F.$$

These quantities separate three layers of error: covariance approximation, operator perturbation of the transport map, and the induced perturbation on the corrected future data. The associated parameter sweeps should vary the target rank, the oversampling parameter, the regularization level λ, and the working precision used in the sketch products.

This second experimental program is also where the balancing-inexactness idea from ??→p.?? becomes directly testable. As the approximation rank grows, randomized truncation error should decrease. If the arithmetic regime is too aggressive, however, a point will be reached at which floating-point error no longer remains hidden beneath the approximation error. The decisive numerical question is therefore not merely whether lower precision is faster, but where the curves begin to separate and how that separation depends on regularization.

The covariance-transport experiments should therefore be organized as explicit tests rather than only as exploratory plots. A concrete first campaign is the following.

1. **Covariance approximation versus rank.** Fix the precision and vary the target rank r to test how quickly $\|C - \widehat{C}\|_2$ decays.
2. **Transport error versus regularization.** Fix the rank and vary λ to test how strongly the inverse-square-root conditioning affects $\|T_\lambda - \widehat{T}_\lambda\|_2$.
3. **Precision threshold sweep.** Compare double, single, and lower precision in the sketch products to identify the point at which floating-point error ceases to be dominated by randomized truncation.
4. **Balancing randomized and arithmetic error.** Increase the rank until the approximation error becomes small enough that arithmetic error is no longer hidden, thereby exposing the regime predicted by the balancing principle.
5. **Climate-shaped synthetic data.** Replace the purely iid Gaussian model by matrices with decaying spectra and block dependence so that the covariance-transport diagnostics are tested in a structure closer to the intended application.

Each of these experiments should be described in the final thesis by four items: the working hypothesis, the matrix or data model, the parameters being swept, and the diagnostics used to decide success or failure. That structure keeps the experimental chapter tied to the theorem statements rather than to ad hoc plotting.

The first concrete realization of this program is in `experiments/benchmark_randmbc.R`. The benchmark is still synthetic, but it already compares three relevant levels: the deterministic full covariance-transport baseline, selected reference methods from the vendored `MBC` package, and the current randomized `randMBC` backend under several rank, oversampling, and precision choices. The present result is not yet a speedup story. On the current synthetic data, the deterministic full baseline and the MBC references remain much more accurate in correlation and future-sample error than the low-rank randomized variants. This is useful evidence rather than a setback. It indicates that the dominant limitation is still the approximation regime itself, not yet the arithmetic regime.

This interpretation also sharpens the mixed-precision question. In the current

Regime	Full baseline	Randomized transport	Mixed precision visible?
Fast spectral decay, moderate λ	good	promising	not dominant
Flat spectrum, small λ	good	unstable	hidden by truncation
Shared basis, adequate λ	good	improved	to test
Independent basis	weaker	unstable	not decisive

Table 1.3: Qualitative regime structure for randomized covariance-transport experiments. "Full baseline" denotes deterministic full covariance transport. "Randomized transport" denotes the low-rank surrogate. "Mixed precision visible?" asks whether reducing precision in the sketch products produces a noticeable change relative to double-precision randomized transport.

benchmark, single and double precision in the randomized path are often very close to one another, while both remain far from the deterministic baseline. That is a sign that truncation and transport perturbation are still larger than the precision effect one is trying to detect. For the later experimental chapter, this means that precision sweeps should be read only after the rank and regularization regime has been pushed into a range where randomized approximation is already credible.

1.9 Regime-Identification Table

The experimental program described above naturally separates into three parameter regimes, each with a distinct limiting factor. Table 1.3 summarizes the qualitative expectations for each regime. The boundaries are not sharp; they depend on the spectral decay of the covariance operators and on the choice of λ. The purpose of the table is to make the regime structure explicit so that later experimental results can be read as regime-identification findings rather than as undifferentiated success-or-failure verdicts.

The first row reflects the most favorable setting: the covariance operators have rapidly decaying spectra, so a modest rank captures most of the variance, and regularization is generous enough to keep the inverse-square-root amplification controlled. The second row is the least favorable: flat spectra and weak regularization make the low-rank surrogate unreliable, and precision effects are invisible because the approximation error already dominates. The third row is the target regime identified by the shared-basis construction in ?? →p. ??: using one reduced coordinate system for both operators stabilizes the transport map even when the raw covariance error is only marginally different. The fourth row serves as the diagnostic baseline for comparison.

Rank r	λ	Mean $\|C - \widehat{C}\|_2$	Mean $\|T_\lambda - \widehat{T}_\lambda\|_2$
5	10^{-2}	0.87	2.35
5	10^{-4}	0.85	25.5
5	10^{-6}	0.84	257.5
10	10^{-2}	0.75	2.09
10	10^{-4}	0.76	23.1
10	10^{-6}	0.72	232.2
15	10^{-2}	0.68	1.66
15	10^{-4}	0.65	17.8
15	10^{-6}	0.67	189.4
20	10^{-2}	0.69	1.28
20	10^{-4}	0.65	13.6
20	10^{-6}	0.72	134.5
25	10^{-2}	0.65	1.13
25	10^{-4}	0.65	8.89
25	10^{-6}	0.66	91.0

Table 1.4: Covariance transport sweep: exponential spectrum, $p = 30$, double precision, Gaussian sketch, 10 trials. All errors are absolute: $\|C - \widehat{C}\|_2$ and $\|T_\lambda - \widehat{T}_\lambda\|_2$. Means over trials; per-trial variability not yet available in post-processed form.

1.10 Covariance Transport Results

The second set of experiments tests the theory from ??→p.?? directly. A covariance operator with exponential spectral decay is generated using `generate_cov_operator` with $p = 30$ and condition number 100. The target rank is swept over $r \in \{5, 10, 15, 20, 25\}$, the regularization parameter over $\lambda \in \{10^{-2}, 10^{-4}, 10^{-6}\}$, and the sketch precision over double and single. Ten independent trials are drawn for each parameter combination.

Table 1.4→p.9 reports the mean covariance error and transport error for double precision. The most important observation is the strong dependence on λ: at $\lambda = 10^{-6}$, the transport error is two orders of magnitude larger than at $\lambda = 10^{-2}$ even though the covariance errors are comparable. This is exactly the inverse-square-root amplification mechanism predicted by ??→p.?? in ??→p.??. The covariance error itself decreases slowly with rank, reflecting the exponential but not extremely steep spectral decay.

Rank r	λ	Double precision	Single precision
5	10^{-2}	2.35	2.38
5	10^{-6}	257.5	259.6
20	10^{-2}	1.28	1.33
20	10^{-6}	134.5	136.6
25	10^{-2}	1.13	1.10
25	10^{-6}	91.0	86.1

Table 1.5: Precision comparison for selected parameter combinations. Mean absolute transport error $\|T_\lambda - \widehat{T}_\lambda\|_2$ over 10 trials. Per-trial variability not yet available.

Precision comparison

Table 1.5→p.10 compares single and double precision at selected parameter combinations. The key finding is that single and double precision produce nearly identical results across all tested regimes. At rank 20 and $\lambda = 10^{-6}$, for example, the transport errors are 134.5 (double) and 136.6 (single). This is not an accident: in all tested cases, the randomized truncation error and the regularization-dominated amplification error are both much larger than the arithmetic precision floor. The floating-point contribution is effectively hidden beneath the approximation error, exactly as the balancing-inexactness principle predicts.

This finding is simultaneously reassuring and limiting. It confirms that single precision is safe in the current approximation regime, but it also means that the present experiments cannot yet detect the precision-visible regime. To observe the crossover, one would need either a higher target rank (so that truncation error falls below the arithmetic floor) or a more aggressive regularization (so that the transport map is not dominated by the inverse-square-root instability).

Figure 1.6→p.11 shows the mean transport error as a function of target rank for each regularization level (double precision). The curves separate dramatically by λ, with the $\lambda = 10^{-2}$ curve remaining near unity while the $\lambda = 10^{-6}$ curve stays above 80 even at rank 25. This confirms that the dominant bottleneck is the inverse-square-root conditioning, not the rank of the approximation.

1.11 Covariance-Transport Experiments: Part II

The second part of the experimental chapter tests the theory from ??→p.?? directly. Instead of tracking only the compressed least-squares residual, one now measures the quality of a randomized low-rank covariance approximation and the perturbation it induces in the regularized transport map. The experiments are organized as explicit tests of the theorem structure:

$$\|T_\lambda - \widehat{T}_\lambda\|_2 \lesssim K_\lambda(C_X, C_Y)\,(\eta_{\mathrm{rand}} + \eta_{\mathrm{fp}}),$$

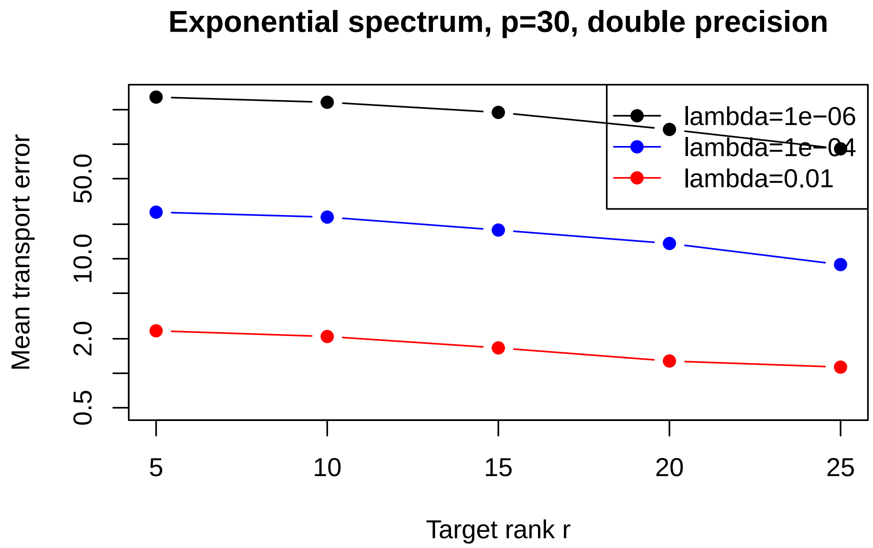

Figure 1.6: Mean absolute transport error versus target rank for three regularization levels (exponential spectrum, $p = 30$, double precision, Gaussian sketch). The vertical axis is logarithmic.

where K_λ is the amplification factor determined by the regularized spectral gaps, η_{rand} bounds the randomized truncation error, and η_{fp} bounds the floating-point perturbation. Each experiment varies one axis while holding the others fixed.

Covariance approximation error versus rank

Hypothesis. For covariance operators with spectral decay, $\|C - \widehat{C}\|_2$ decreases as the target rank increases, with a rate governed by the spectral tail beyond the chosen rank.

Setup. Fix $\lambda = 10^{-2}$ and double precision. Generate covariance operators from each spectrum family (exponential, linear, gap, climate-shaped) using `generate_cov_operator` and `generate_climate_cov`. Vary the target rank from small values up to p.

Diagnostics. $\|C - \widehat{C}\|_2$ (source and target separately), residual norm estimate from the range finder, and runtime.

Transport error versus regularization

Hypothesis. For fixed rank, transport error is controlled by λ through the inverse-square-root amplification factor. Too-small λ leads to instability; too-large λ changes the transported operator.

Setup. Fix rank at a moderately large value and double precision. Vary λ from 10^{-6} to 10^{-1}.

Diagnostics. $\|T_\lambda - \widehat{T}_\lambda\|_2$, minimum eigenvalue of the regularized covariance, condition number estimate, and Pearson correlation error of the corrected future sample.

Precision threshold sweep

Hypothesis. In the regime where randomized truncation error dominates floating-point error, single precision in the sketch products produces nearly identical results to double precision. A crossover point appears when the rank is large enough that truncation error falls below the arithmetic error floor.

Setup. Fix rank and λ. Compare double, single, and bfloat16 simulation in the sketch products via `precision_model`.

Diagnostics. Transport error at each precision, ratio $\eta_{\mathrm{fp}}/\eta_{\mathrm{rand}}$ estimated from `sketch_forward_bound`, and the `is_fp_hidden` flag.

Shared versus independent basis

Hypothesis. The shared-basis construction stabilizes the transport map by eliminating the coordinate mismatch between source and target reduced models, even when the raw covariance error is comparable.

Setup. For the same rank, λ, and precision, compare the independent-subspace and shared-basis variants from ??→p.??.

Diagnostics. Transport error, future-sample perturbation $\|Z_f T_\lambda - Z_f \widehat{T}_\lambda\|_F$, and the same statistical diagnostics as in Section 1.12→p.12.

Balancing randomized and arithmetic error

Hypothesis. The balanced regime $\eta_{\mathrm{fp}} \lesssim \eta_{\mathrm{rand}}$ is achievable and detectable: increasing rank reduces η_{rand} until a crossover where further rank increase reveals floating-point error.

Setup. Fix λ and single-precision sketch. Vary the rank from small to large. Track both error sources.

Diagnostics. Estimated η_{rand} (from range-finder residual), estimated η_{fp} (from `sketch_forward_bound`), their ratio, and the total transport error.

1.12 Application-Facing Diagnostics

Once the covariance-transport experiments are connected to a climate-shaped data pipeline, the numerical diagnostics should be accompanied by statistical ones. At a minimum, one should compare the corrected output against the target dependence structure through marginal quantile diagnostics, correlation diagnostics, and multivariate summaries of dependence error. Useful quantities include Pearson and Spearman correlation error, spatial and temporal autocorrelation error, and an energy-score-style multivariate discrepancy. The point is not to replace the operator-norm viewpoint, but to relate it to the application metrics by which a post-processing method would ultimately be judged.

This distinction is important for interpretation. A method may show a visible operator perturbation while still producing acceptable application-level dependence corrections, or conversely may appear cheap and accurate in low-rank terms while failing to preserve the dependence features that matter downstream. The final experimental design should therefore keep both scales visible: the numerical scale of randomized and floating-point perturbation, and the application scale of multivariate climate diagnostics.